

Erdős Problem #774

EP-774

L1 - TRACTABLE

PARTIALLY SOLVED

Number Theory

ORIGINAL PROBLEM STATEMENT

We call $A \subset \mathbb{N}$ dissociated if $\sum_{n \in X} n \neq \sum_{m \in Y} m$ for all finite $X, Y \subset A$ with $X \neq Y$.

Let $A \subset \mathbb{N}$ be an infinite set. We call A proportionately dissociated if every finite $B \subset A$ contains a dissociated set of size $\gg |B|$.

Is every proportionately dissociated set the union of a finite number of dissociated sets?

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Archive: <https://www.unsignedmath.com/problems?difficulty=1>

Dataset: <https://huggingface.co/datasets/ulamai/UnsignedMath>